

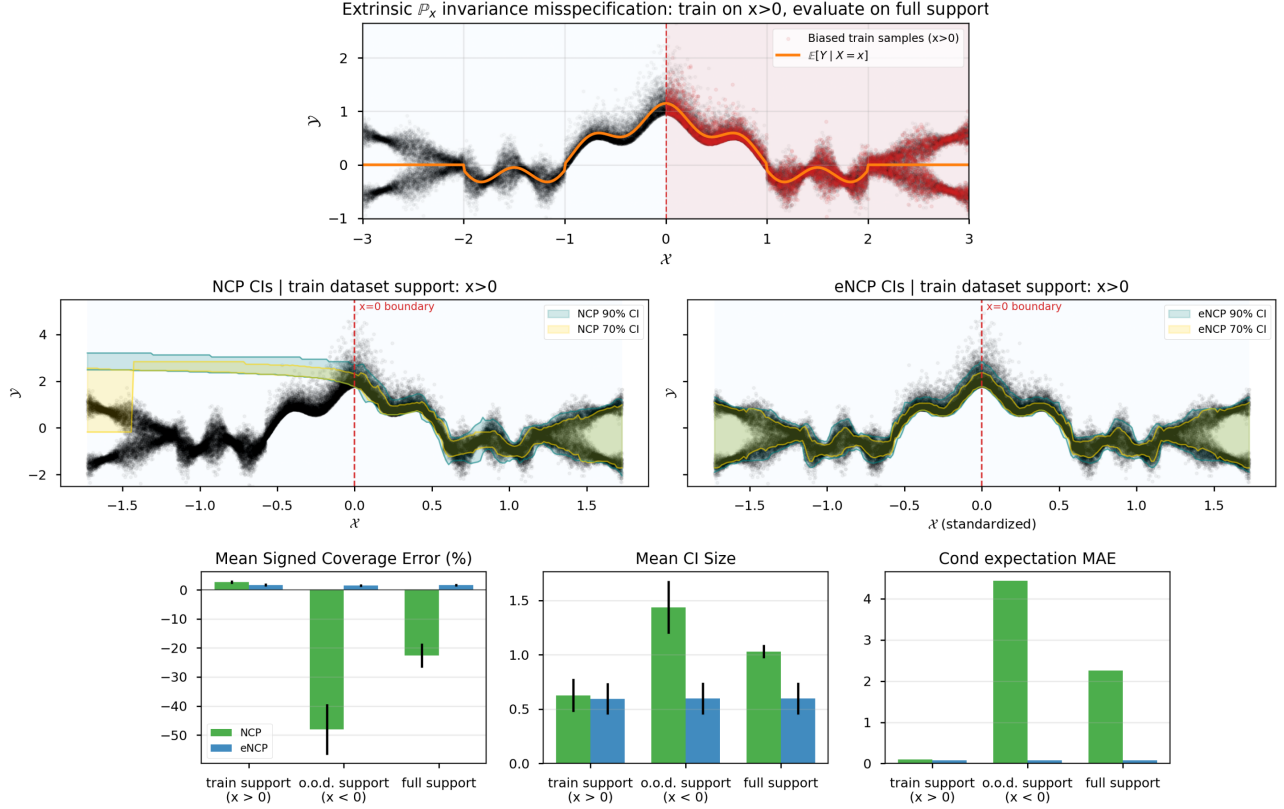


Figure A1. Top: Illustration of extrinsic symmetry misspecification, where the train/validation/test support is biased toward the coset subspace  $\mathbf{x} > 0$ , violating the assumed  $\mathbb{C}_2$ -invariance of  $P_{\mathbf{x}}$ . Middle: Predicted confidence intervals at 90% and 70% coverage for NCP and eNCP under extrinsic symmetry misspecification. Bottom: Mean coverage error, confidence-interval volume, and conditional-expectation Maximum Absolute Error (MAE) for NCP and eNCP, evaluated on the biased support  $\mathbf{x} > 0$ , the out-of-distribution (o.o.d.) support  $\mathbf{x} < 0$ , and the full support  $\mathcal{X} \equiv \mathbb{R}$ .

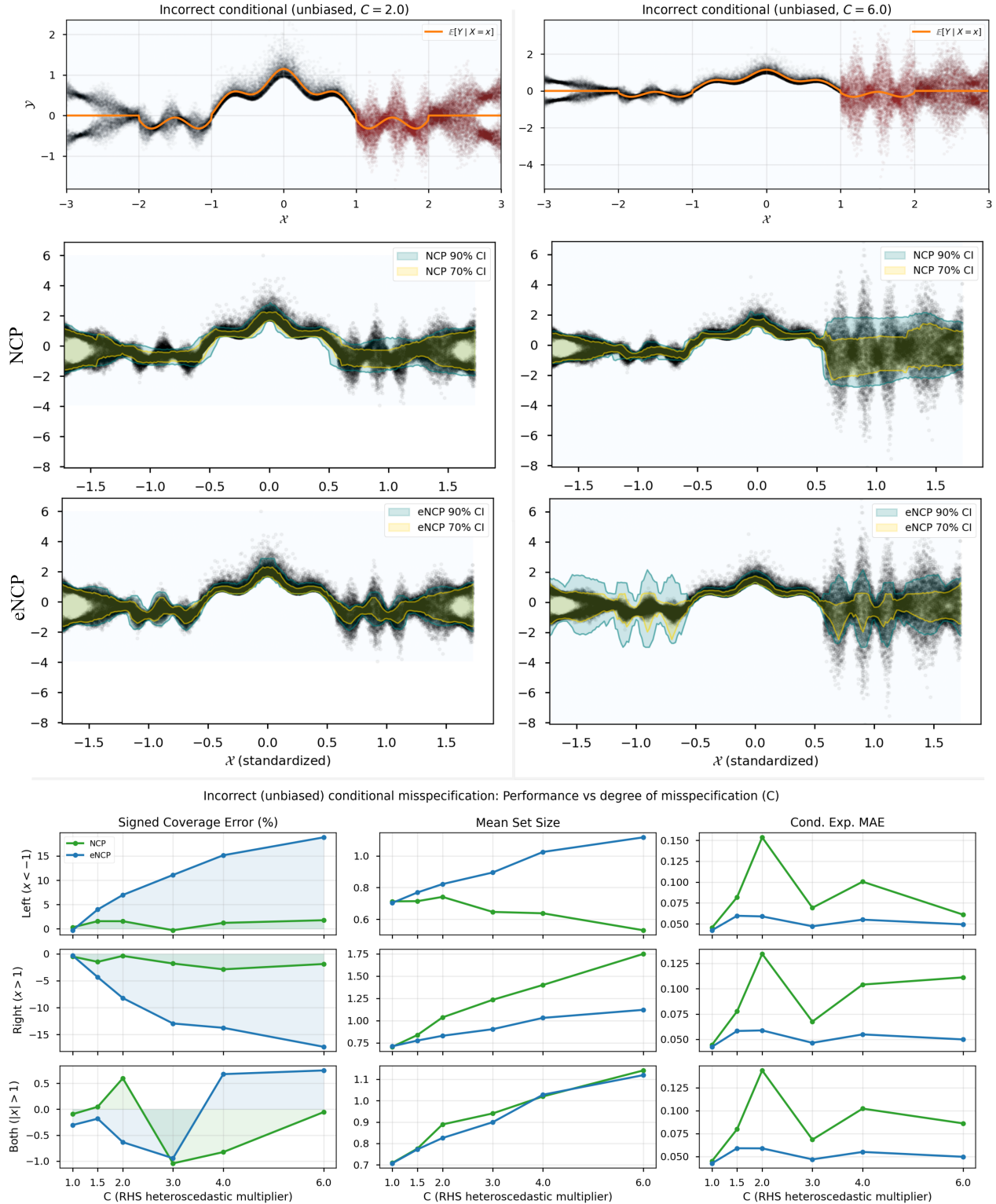


Figure A2. First row: Illustration of incorrect (unbiased) conditional misspecification. The  $\mathbb{C}_2$ -invariance of  $P_{Y|X}$  is violated on  $|x| > 1$  by increasing the heteroscedastic noise scale with  $C \in [1.0, 6.0]$  only on the right-coset subspace  $x > 1$ . This misspecification is termed unbiased because the conditional expectation  $\mathbb{E}[y | x = \cdot]$  remains  $\mathbb{C}_2$ -equivariant. Second and third rows: Predicted confidence intervals at 90% and 70% coverage for NCP and eNCP with  $C = 2.0$  and  $C = 6.0$  under this incorrect unbiased conditional misspecification. Bottom: Mean (signed) coverage error, confidence-interval volume, and conditional-expectation Maximum Absolute Error (MAE) for NCP and eNCP, evaluated on the misspecified supports  $x > 1$  and  $x < -1$ , and on the full support  $\mathcal{X} \equiv \mathbb{R}$ , across noise scales  $C \in [1.0, 6.0]$ .

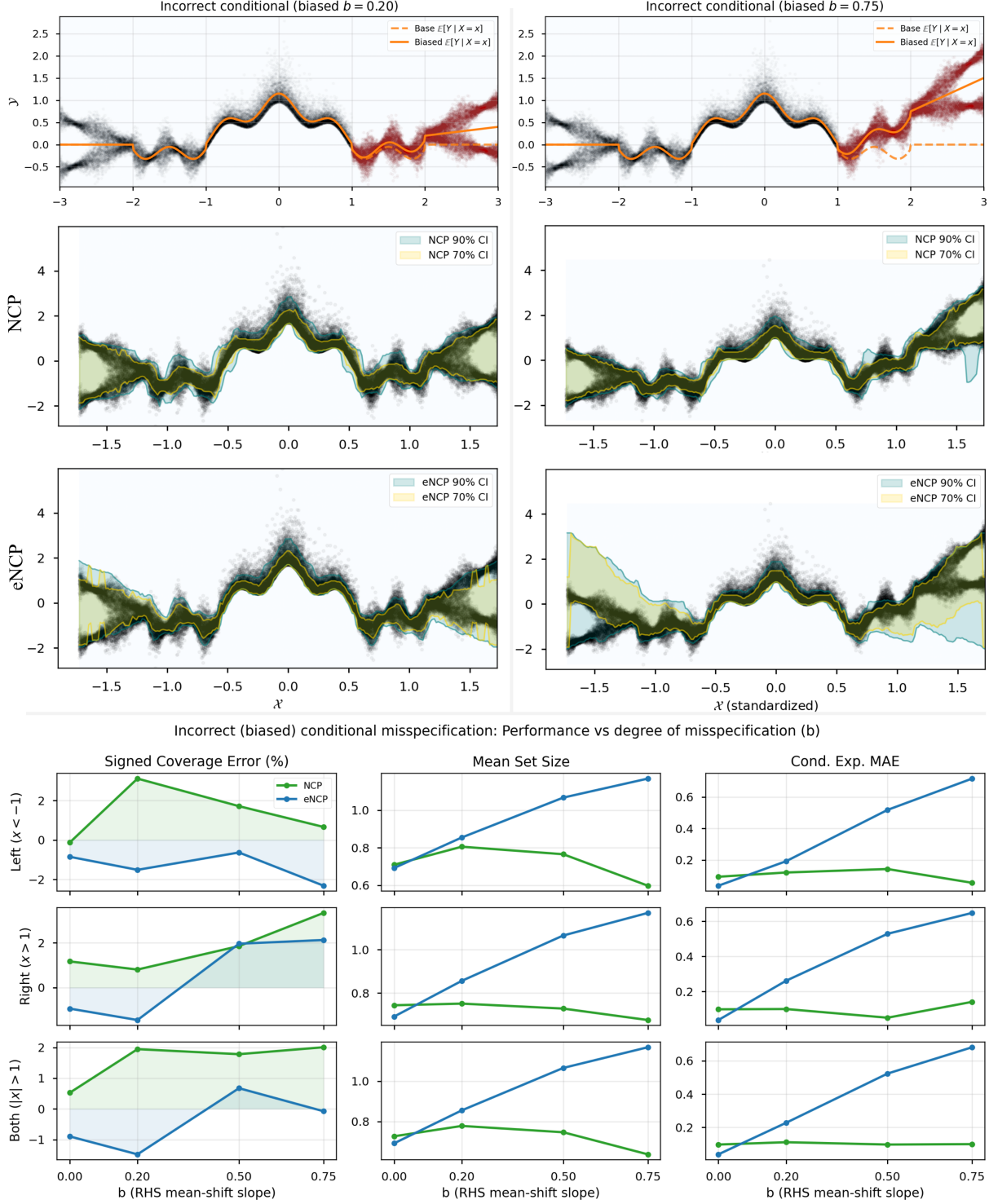


Figure A3. First row: Illustration of incorrect (biased) conditional misspecification. The  $\mathcal{C}_2$ -invariance of  $P_{y|x}$  is violated on  $|x| > 1$  by introducing a linear bias with slope  $b \in [0.0, 0.75]$  on the right-coset subspace  $x > 1$ . This misspecification is termed biased because the conditional expectation  $\mathbb{E}[y | x = \cdot]$  is no longer  $\mathcal{C}_2$ -equivariant. Second and third rows: Predicted confidence intervals at 90% and 70% coverage for NCP and eNCP with  $b = 0.2$  and  $b = 0.75$  under this incorrect biased conditional misspecification. Bottom: Mean (signed) coverage error, confidence-interval volume, and conditional-expectation Maximum Absolute Error (MAE) for NCP and eNCP, evaluated on the misspecified supports  $x > 1$  and  $x < -1$ , and on the full support  $\mathcal{X} \equiv \mathbb{R}$ , across bias slopes  $b \in [0.0, 0.75]$ .